

VLM-DexSafeDagger: VLM-Guided Intervention Policy for Safety-Critical Visuomotor Skill Distillation

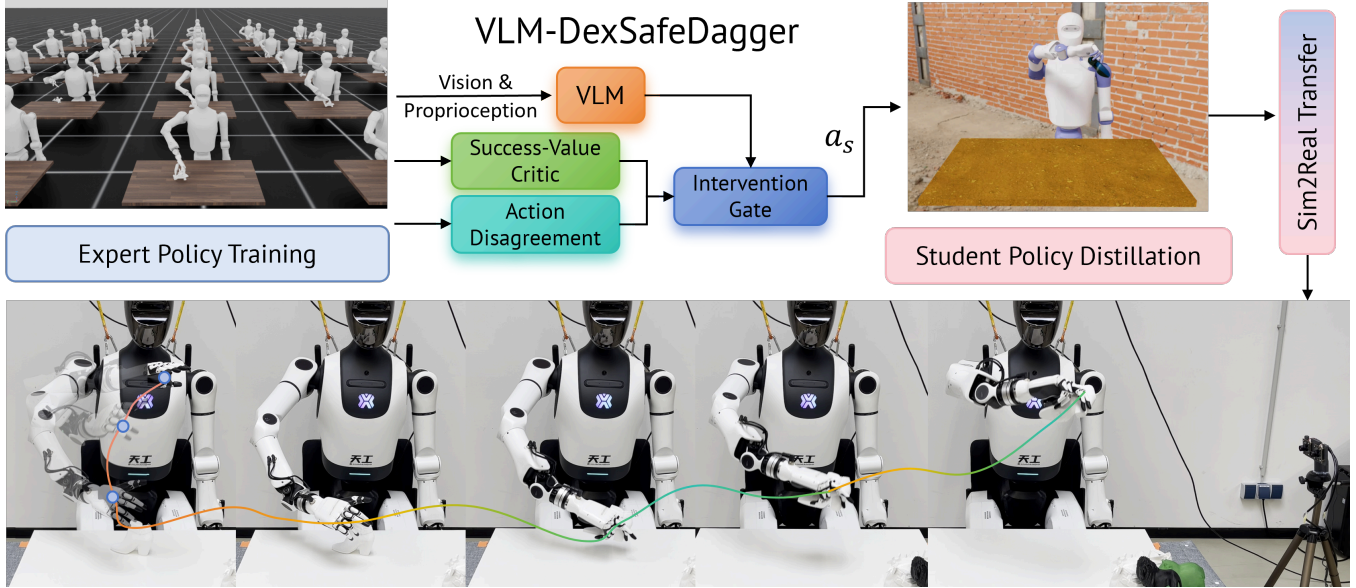


Fig. 1: **VLM-DexSafeDagger overview.** We propose a teacher-intervened online distillation pipeline that condenses teacher skills into a deployable RGB-based student policy. The unsafe exploration rate is substantially suppressed in the early training phase via a proposed VLM-guided intervention policy that combines teacher–student action disagreement with a Bellman success-value critic trained on safe-lift-success labels, while a vision-language model maintains an evolving, safety-aware understanding of the scene and periodically adjusts the intervention policy during online distillation.

Abstract—Learning dexterous manipulation is challenging due to high-dimensional action spaces and complex contact dynamics. In addition, safety-unaware exploration during learning can lead to hardware damage and unstable or even hazardous contact events, posing a significant barrier to real-world deployment. To address this challenge, we propose VLM-DexSafeDagger, a safety-aware online skill-distillation framework that transfers reinforcement learning (RL) teacher policies to a vision-based student policy. VLM-DexSafeDagger combines an ℓ_2 teacher–student disagreement gate with a learned Bellman success-value critic that estimates the prospective safe-lift-success value of the current student state–action pair. A vision-language model periodically adjusts the intervention policy parameterized by disagreement limit and acceptable predicted success rate. This design enables safer exploration during online interactive imitation learning using high-level visual context. Experiments on grasp-and-lift tasks show that VLM-DexSafeDagger significantly improves training safety, reducing unsafe episodes by up to 80% during the training period compared with vanilla DAgger-based distillation. Meanwhile, the distilled student policy achieves comparable lift success rates and demonstrates sufficient robustness for real-world robotic deployment.

I. INTRODUCTION

Dexterous robotic hands typically possess a large number of degrees of freedom (DoFs) and exhibit complex contact

dynamics, enabling them to perform fine-grained and diverse manipulation behaviors, such as multi-finger grasping [1], in-hand reorientation [2], [3], and precise control [4]. However, the high-dimensional action space and intricate interaction dynamics substantially increase the difficulty of control and policy learning. In recent years, reinforcement learning (RL)-based approaches [3], [5]–[7] have achieved remarkable progress in dexterous grasping and manipulation tasks. Leveraging large-scale parallel simulation and high-fidelity physics engines, RL enables the automatic learning of complex control policies in simulated environments without the need for manually designed controllers. These methods have demonstrated strong generalization capabilities across multi-object and multi-task scenarios.

Despite these advances, existing RL frameworks primarily optimize for task success rates and rarely incorporate explicit modeling or constraints related to safety-critical behaviors. During training, common failure modes include severe collisions between the hand and the environment, unintended object ejection due to excessive contact forces, and loss of stable hand posture [8], [9]. Such unsafe behaviors not only reduce training efficiency but also introduce potential risks during real-world deployment. Therefore, how to reduce

unsafe exploration while maintaining learning efficiency remains a critical challenge in dexterous manipulation.

To mitigate the performance gap between simulation and real-world deployment, skill distillation has become a widely adopted paradigm in recent years. The core idea is to first train a high-performing expert policy in simulation using privileged state information, and then distill its decision-making behavior into a student policy that relies solely on visual observations or partially observable inputs. This process enables modality transfer from privileged state information to perception-based inputs, facilitating efficient real-world deployment. Skill distillation reduces reliance on accurate state estimation and allows knowledge integration from multiple teacher policies, thereby substantially improving execution stability and generalization performance in real-world environments. Consequently, it has gained increasing attention in dexterous grasping and complex manipulation tasks. However, existing distillation methods [10], [11] typically rely on behavior cloning or the standard DAgger framework for data aggregation. Their primary objective is to mitigate performance degradation caused by distribution shift, rather than explicitly enforcing safety constraints during student policy learning. During distillation, the student policy may deviate from the teacher’s state-action distribution and enter regions that are either insufficiently covered by the teacher or inherently high-risk, leading to unsafe behaviors such as severe collisions or unstable hand configurations. Therefore, introducing explicit safety mechanisms into the distillation process—so that high-risk exploration can be avoided without sacrificing learning efficiency—remains an important and largely underexplored problem. Existing safety-aware distillation approaches, such as SafeDAgger [12], have primarily focused on autonomous driving scenarios and have not been systematically studied in the context of dexterous manipulation.

The contributions of this work are summarized as follows:

- We propose VLM-DexSafeDAgger, a VLM-guided, safety-sensitive framework for distilling RL-acquired teacher policies into a single RGB-based student policy. To the best of our knowledge, this is the first attempt in dexterous manipulation to incorporate a mixed intervention policy combining teacher-student action disagreement with a Bellman success-value critic, while the VLM reasons about scene-level safety from rollout statistics and representative visual observations to regulate the intervention gate.
- We propose a safety-aware RL reward design that explicitly accounts for risky events in arm-hand control scenarios, including harmful collisions, object escape, palm flipping, and the hand moving too far from the workspace, thereby encouraging safer exploration during teacher training.
- We empirically demonstrate that VLM-DexSafeDAgger, which combines disagreement, success-value prediction, and VLM-guided intervention policy adaptation, substantially reduces unsafe exploration compared with Vanilla DAgger, achieving an approximately **80%** rel-

ative reduction in the unsafe-episode rate during early training. It also reduces the average unsafe-episode rate by **7.5 percentage points** relative to disagreement-only SafeDAgger [12]. Ablation results further demonstrate the effectiveness of VLM-guided adaptation across different visual contexts and unsafe scenarios: VLM-DexSafeDAgger consistently achieves lower unsafe-episode rates than DexSafeDAgger, with an average reduction of **4 percentage points**.

II. RELATED WORK

Skill Distillation. Skill distillation transfers behavior from a high-capacity teacher to a compact student and has been widely adopted in both computer vision and robotics. In robotics, the teacher can leverage privileged state and simulator access during training, while the student is distilled to operate from on-board observations (e.g., RGB). DexTrAH-RGB [11] distills a privileged RL teacher into an RGB-based policy via DAgger-based imitation learning, while UniGraspTransformer [13] distills offline trajectories from object-specific teachers into a unified Transformer policy.

Interactive Imitation Learning. Many imitation learning methods operate offline, using a fixed dataset of expert demonstrations to match the expert action distribution [14], [15]. In sequential decision-making, however, distributional shift induces covariate shift and compounding errors. Interactive imitation learning [16] mitigates this issue by collecting supervision online along trajectories visited by the learner.

DAgger [10] addresses covariate shift by iteratively rolling out the learner and relabeling visited states with an expert, thereby aggregating data under the learner-induced state distribution. SafeDAgger [12] extends DAgger with a teacher-intervened rollout policy that takes over in high-risk states to reduce unsafe exploration. Recent variants [17], [18] extend interactive imitation learning to robotic manipulation, yet often rely on human supervision. Fully autonomous expert relabeling during online distillation remains rare in robotic manipulation, especially for dexterous, contact-rich tasks.

Vision-Language-Guided Robot Learning. Visual and semantic information is widely recognized as a strong cue for embodied agents to interpret task specifications and reason about unstructured environments [19], [20]. One line of work uses VLMs to generate demonstrations or supervisory signals for offline policy construction, such as Manipulate-Anything [21] and AHA [22]. Complementary approaches, including RACER [23] and FOREWARN [24], employ VLMs to guide risk prediction, provide recovery guidance for unsafe situations, or govern action selection during online policy learning or real-time execution.

III. PROBLEM STATEMENT AND METHODS

A. Problem Formulation

We consider online interactive imitation learning with teacher intervention in a partially observable Markov decision process (POMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, \Omega, r, \gamma)$, with privileged teacher state $s_t \in \mathcal{S}$, deployable student observation $o_t \in \mathcal{O}$ (RGB), action $a_t \in \mathcal{A}$, observation model $o_t \sim \Omega(\cdot \mid s_t)$,

Algorithm 1 VLM-DexSafeDagger

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1: Train the privileged teacher policy  $\pi_T$  via RL
2: Collect a warm-start dataset  $\mathcal{D}_0$  under teacher control
3: Pretrain the Bellman critic  $\hat{Q}_{\text{succ}}$  on  $\mathcal{D}_0$ 
4: Initialize the intervention policy parameters  $\tau^{(0)} = (\tau_\delta^{(0)}, \tau_Q^{(0)})$ 

5: Initialize the visual buffer  $\mathcal{B}_V$ 
6: for iteration  $k = 1 \dots N$  do
7:   Observe the teacher state  $s_t$  and student observation  $o_t$ 
8:   Compute the actions  $a_t^T \sim \pi_T(\cdot | s_t)$  and  $a_t^S \sim \pi_S(\cdot | o_t)$ 
9:   Compute the action disagreement  $\delta_t = \|a_t^S - a_t^T\|_2$ 
10:  Predict the safe-lift-success value  $\hat{Q}_{\text{succ}}(s_t, a_t^S)$ 
11:  Evaluate the intervention gate
12:    $g_t \leftarrow \mathbb{I}[\delta_t > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q]$ 
13:  Execute the gated action  $a_t = g_t a_t^T + (1 - g_t) a_t^S$ 
14:  Store the transition in the critic replay buffer  $\mathcal{D}$ 
15:  Update  $\hat{Q}_{\text{succ}}$ ; if  $g_t = 1$ , update  $\pi_S$ 
16:  Add representative visual states to  $\mathcal{B}_V$ 
17:  if an intervention policy update is due then
18:    Summarize rollout context  $\mathcal{C}_k$  and sample images  $\mathcal{V}_k$ 
19:    Query the VLM for the shift  $\Delta\tau^{(k)}$  given  $(\mathcal{C}_k, \mathcal{V}_k)$ 
20:    Form the candidate  $\tilde{\tau}^{(k)} = \tau^{(k)} + \Delta\tau^{(k)}$ 
21:    Clip and smooth  $\tilde{\tau}^{(k)}$  to obtain  $\tau^{(k+1)}$ 
22:  end if
23: end for

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TABLE I: Reward components used for teacher policy training.

Term	Name	Purpose
r_{dist}	Hand distance	Guides the hand toward the object.
r_{curl}	Finger curl	Encourages a grasp-ready finger posture.
r_{align}	Palm alignment	Aligns the palm to a palm-down pose.
r_{contact}	Contact bonus	Rewards stable thumb–finger contact.
r_{near}	Stay-close	Keeps the hand near the object after reaching.
r_{goal}	Goal tracking	Guides the grasped object toward the goal.
r_{lift}	Lift	Rewards lifting after stable contact.
r_{action}	Action smoothness	Penalizes abrupt action changes.

The teacher is trained in simulation using an asymmetric actor–critic setup. Details of the actor and critic observation spaces are provided in the Appendix. The actor receives noisy and partial proprioceptive observations that approximate realistic sensing, while the critic is provided with the privileged full state of both the robot and object, available only in simulation.

Episodes terminate at

$$T = \min \{t \mid \mathbb{I}_{\text{unsafe}}(s_t) = 1 \text{ or } t \geq T_{\text{max}}\}, \quad (3)$$

where $T_{\text{max}} = 10$ s is the timeout horizon. Early termination conditions include palm flip, harmful collision, object escape, and hand-too-far (see Sec. A), and are treated as unsafe events for both teacher training and subsequent intervention decisions.

D. Bellman Success-Value Critic

ℓ_2 action disagreement alone is insufficient as an intervention signal and may miss small action deviations that could cause unstable grasps, collisions, or object escape. We therefore learn a Bellman critic, trained with safe-lift-success labels, that estimates the probability of safe lift success for the current student state–action pair and provides a prospective signal complementary to action disagreement.

The Bellman critic evaluates a multimodal state–action pair (s_t, a_t^S) , where s_t combines robot proprioception and embeddings from a visual encoder. The encoder is trained end-to-end during skill distillation and regularized by the auxiliary object-position prediction loss in Sec. G. The twin critics Q_1 and Q_2 are ReLU MLPs with hidden dimensions [256, 128]. Each critic’s logit $z_i(s, a)$ is mapped to a success probability by $Q_i(s, a) = \sigma(z_i(s, a)/T)$, with calibration temperature $T = 4$. We use the conservative estimate $\hat{Q}_{\text{succ}}(s, a) = \min_{i \in \{1, 2\}} Q_i(s, a)$; lower values indicate higher prospective risk.

The critics are pretrained on teacher transitions and updated online from a ring replay buffer. The label $u_t \in \{0, 1\}$ indicates safe lift success: it is 1 only when the lift criterion is satisfied without an unsafe event. Training uses the SARSA-style target

$$y_t = \text{clip} \left(u_t + \gamma(1 - d_t^{\text{eff}}) \min_{i \in \{1, 2\}} Q_i^{\text{targ}}(s_{t+1}, a_{t+1}), 0, 1 \right), \quad (4)$$

where d_t is the termination flag, h_t indicates whether a_{t+1} is valid, $d_t^{\text{eff}} = \max(d_t, 1 - h_t)$, and Q_i^{targ} are slowly updated target networks. Both critics minimize the mean-squared error (MSE) between their predictions and y_t ; the minimum across critics reduces overestimation.

E. Intervention Policy

The intervention policy uses a deterministic two-channel gate. The first channel measures teacher–student disagreement, $\delta_t := \|a_t^S - a_t^T\|_2$, where $a_t^T \sim \pi_T(\cdot | s_t)$ and $a_t^S \sim \pi_S(\cdot | o_t)$. The second channel evaluates prospective task success through $\hat{Q}_{\text{succ}}(s_t, a_t^S)$. The intervention policy is parameterized by $\tau^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, where $\tau_\delta^{(k)}$ is the maximum admissible disagreement and $\tau_Q^{(k)}$ is the minimum acceptable predicted success.

These parameters define the deterministic intervention gate:

$$g_t = \mathbb{I}[\hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q^{(k)} \vee \delta_t > \tau_\delta^{(k)}]. \quad (5)$$

Thus, either large disagreement or low predicted success triggers teacher takeover. When the two signals conflict, the conservative OR rule applies: low teacher–student disagreement does not suppress intervention if the critic predicts low student success, and a favorable critic prediction does not suppress intervention if the action disagreement is large. The student acts only when both conditions are satisfied. Teacher takeover is treated as empirically safer rather than as an absolute safety guarantee, and the resulting safety performance remains bounded by the teacher policy’s own performance ceiling. Decreasing τ_δ or increasing τ_Q makes the intervention policy more conservative; the opposite shift increases student autonomy.

We empirically initialize $\tau_\delta^{(0)}$ and $\tau_Q^{(0)}$ from the 90th percentile of warm-start disagreement and the 10th percentile of predicted success over successful teacher rollouts, respectively. These values define the initial operating point $\tau^{(0)}$.

F. VLM-Guided Intervention Policy Adaptation

During online distillation, the student policy, visited-state distribution, and Bellman critic’s calibration co-evolve, and

the dominant failure modes may also change. A stationary intervention policy therefore cannot maintain a consistent safety–autonomy trade-off. We periodically query a VLM to guide the displacement of the intervention policy’s operating point. At iteration k , the VLM receives rollout statistics $\mathcal{C}_k$, including disagreement (δ_t), predicted success ($\hat{Q}_{\text{succ}}(s_t, a_t^S)$), intervention rate (β), and unsafe-episode rate (UnsafeRateAvg). It also receives a representative image batch $\mathcal{V}_k$ sampled from the visual buffer $\mathcal{B}_V$ to cover high-disagreement, low-success, and unsafe states. From this context, it proposes an operating-point displacement $\Delta\tau^{(k)} = (\Delta\tau_\delta^{(k)}, \Delta\tau_Q^{(k)})$, producing the candidate operating point $\tilde{\tau}^{(k)} = \tau^{(k)} + \Delta\tau^{(k)}$.

Each candidate policy parameter is clipped around its initial value and exponentially smoothed:

$$\tau_x^{(k+1)} = (1 - \alpha)\tau_x^{(k)} + \alpha \text{clip}\left(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\right), \\ x \in \{\delta, Q\}. \quad (6)$$

Here, α controls the adaptation rate, and $c_x^{\min}$ and $c_x^{\max}$ define the admissible calibration region around $\tau_x^{(0)}$.

G. Student Skill Distillation

Student learning is posed as supervised imitation from teacher-labeled transitions collected under the gated rollout policy π_G . The student observes RGB images (two stereo views) together with robot proprioception.

We optimize a combined objective $\mathcal{L} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{aux}}$, where the auxiliary loss predicts object position to regularize the visual encoder, $\mathcal{L}_{\text{aux}} = \|\hat{\mathbf{x}}_{\text{obj}} - \mathbf{x}_{\text{obj}}\|_2$.

For action imitation, we use a weighted L_2 loss, $\mathcal{L}_{\text{action}} = \|\mathbf{a}_S - \mathbf{a}_T\|_W^2 = (\mathbf{a}_S - \mathbf{a}_T)^\top W (\mathbf{a}_S - \mathbf{a}_T)$, with $W = \text{diag}(w_1, \dots, w_d)$. We set $w_j = \sigma_{0,j}^{-2}$ using a reference scale $\sigma_{0,j}$ (e.g., the teacher action standard deviation on dimension j), yielding $\mathcal{L}_{\text{action}} = \sum_{j=1}^d (a_{S,j} - a_{T,j})^2 / \sigma_{0,j}^2$.

Training proceeds in two stages. (i) *Warm-start*: we collect an offline dataset (e.g., 100k transitions) under full teacher control ($g_t \equiv 1$) to pretrain the Bellman critic. (ii) *Online distillation*: we perform interactive rollouts under π_G and query teacher labels on intervened steps. The student is updated online using the current-step teacher label only; no replay buffer or offline dataset is maintained for student updating. The design reduces the effect of stale supervision and stabilizes online imitation updates.

IV. EXPERIMENT

A. Evaluation Metrics

We designed the following metrics to evaluate both teacher and student policy performance. Let $\tau \in \{1, \dots, N\}$ index evaluation episodes and $t \in \{0, \dots, T_\tau\}$ index timesteps. Unless otherwise stated, we run the evaluation episodes with a horizon of 10s.

a) Lift success (ever-lifted with hold).

Let h_t^τ be the object height above the table and H a lift threshold. An episode is considered successful if the object remains above H for at least N_{hold} consecutive steps. We

define $\mathbb{I}_{\text{lift}}^\tau = \mathbb{I}[\exists t : \forall k \in \{0, \dots, N_{\text{hold}} - 1\}, h_{t+k}^\tau \geq H]$ and report: $\text{LiftSucc} = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_{\text{lift}}^\tau$.

b) Unsafe episode rate.

As stated above, an episode may terminate early due to one of the following conditions: palm flipped ($\mathbb{I}_{\text{flip}}^\tau$), harmful collision ($\mathbb{I}_{\text{coll}}^\tau$), object escape ($\mathbb{I}_{\text{obj}}^\tau$), or hand too far ($\mathbb{I}_{\text{hf}}^\tau$). Let $\mathcal{M} = \{\text{coll}, \text{obj}, \text{flip}, \text{hf}\}$ denote the set of termination modes, and let $\mathbb{I}_m^\tau \in \{0, 1\}$ indicate whether mode $m \in \mathcal{M}$ occurs in episode τ . An episode is unsafe if any termination mode is triggered: $\mathbb{I}_{\text{unsafe}}^\tau = \mathbb{I}[\bigvee_{m \in \mathcal{M}} \mathbb{I}_m^\tau = 1]$. We report: $\text{UnsafeRateAvg} = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_{\text{unsafe}}^\tau$.

c) Failure-mode decomposition.

Each episode is assigned a single terminal mode $m^\tau \in \mathcal{M}$ from the set of unsafe events (collision, object escape, palm flip, hand-too-far) as well as timeout/success where applicable. We define $\mathbb{I}_m^\tau = \mathbb{I}[m^\tau = m]$ and report $\text{FailureModeRate}(m) = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_m^\tau, m \in \mathcal{M}$.

When multiple unsafe conditions are triggered in the same timestep, we record the first triggered termination condition (using a fixed priority order) as the episode’s failure mode.

B. Simulation Experiment Setup

Simulation experiments use single-arm control on a Tian-gong 2 Pro humanoid equipped with InspireHand RH56DfX; the action space is 13-D (7 arm DoF + 6 hand DoF). The RGB student additionally receives two 320×240 camera streams at 30 fps, encoded by a Siamese ResNet-18 visual backbone with view fusion into a 128-D feature appended to the actor observation. Teacher training and student distillation are conducted in Isaac Sim 5.0.0 / Isaac Lab 2.2.1 on a single RTX 5090 GPU.

We evaluate 40 objects from the Omniverse Asset Library and train one object-specific teacher for each object. Each teacher is trained for 12,000 episodes (3.5 h), totaling 480,000 teacher-training episodes and approximately 140 GPU-hours. Across the 40 teachers, lift-success rates range from 72–100%, and unsafe rates range from 5–30%; Table II reports a representative four-object subset. During distillation, we randomize lighting intensity, direction, background textures, object material, and table appearance following DexTraH-RGB-style domain randomization. The joint student is distilled across all 40 objects for 100,000 episodes in 2.5 h.

We use GPT-5.5 with high reasoning effort as the VLM. We issue the first VLM query after 2,000 episodes and subsequently query it every 10,000 distillation episodes, resulting in 10 VLM calls over the full 100,000-episode training run and approximately 70,000 tokens of total usage per run.

C. Ablation Studies

We perform the following ablation studies to validate the design choices in VLM-DexSafeDagger and answer four questions: (i) **Does our customized RL recipe produce stronger teachers than the DexTraH-RGB-style training recipe?** (ii) **Does the intervention mechanism in DexSafeDagger enable safer exploration during policy distillation?** (iii) **Does augmenting action disagreement**

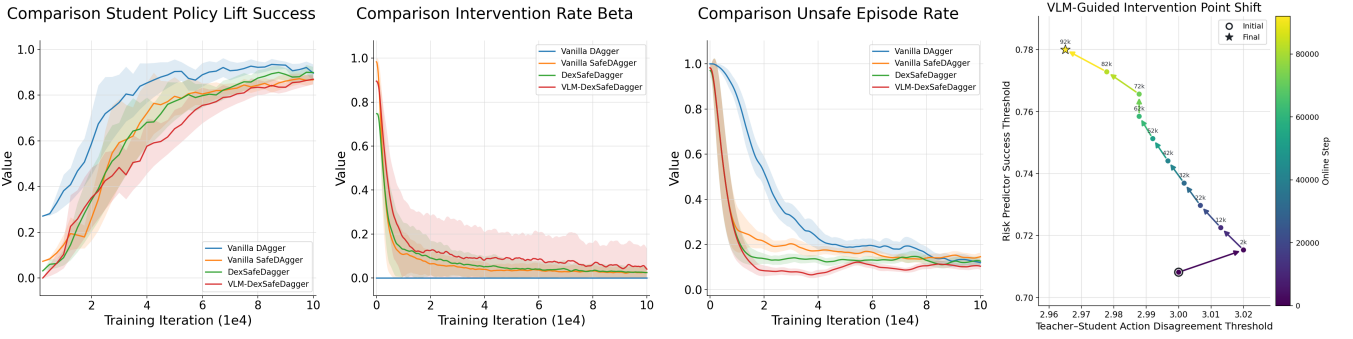


Fig. 3: **Ablation studies.** The first three panels compare four distillation variants in terms of student lift success, teacher intervention rate β , and unsafe episode rate, respectively, all plotted over training iterations. The rightmost panel shows the VLM-guided trajectory of the intervention operating point over online training. Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to vanilla DAgger (blue), vanilla SafeDagger (orange), DexSafeDagger (green), and VLM-DexSafeDagger (red).

TABLE II: Teacher policy performance on four representative objects from the 40-object training set (200 evaluation episodes per object).

Metric (%)	WindCart	Boot	ToolBox	Cow
Lift Success	93.5	100.0	88.5	87.0
Unsafe Rate Avg	6.5	13.5	13.5	13.5
Harmful collision	0.0	0.0	6.5	0.0
Object escape	6.5	6.5	6.5	13.5
Palm flipped	0.0	6.5	0.0	0.0
Hand too far	0.0	0.0	0.0	0.0

with success-value prediction improve the intervention signal? (iv) Is visual-language reasoning necessary for adapting the intervention boundary?

(a) DexTrAH-RGB RL + Vanilla DAgger. The teacher is trained using the DexTrAH-RGB RL recipe and then distilled into a student policy using supervision only, with no teacher intervention during rollouts ($\beta = 0$).

(b) Our RL + Vanilla DAgger. This setting matches (a), except that our RL recipe replaces the DexTrAH-RGB teacher-training recipe.

(c) Our RL + Vanilla SafeDagger. The teacher training follows the recipe in (b), but we replace Vanilla DAgger with a disagreement-gated SafeDagger baseline. The teacher intervenes only when $\|a_t^T - a_t^S\|_2$ exceeds a fixed threshold.

(d) DexTrAH-RGB RL + DexSafeDagger. We use the teacher-training recipe from (a) and the intervention baseline from (c), extending it with a Bellman success-value critic under a stationary intervention policy. Teacher intervention is triggered by either large teacher-student disagreement or low predicted success.

(e) Our RL + DexSafeDagger. Following the recipe in (d), we replace the teacher-training pipeline with our customized RL recipe.

(f) Our RL + VLM-DexSafeDagger. This setting extends (e) by replacing the stationary intervention policy with the proposed VLM tuner, which periodically calibrates the disagreement-risk intervention policy using rollout statistics

TABLE III: Performance of ablations (a)–(h) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
Ablation (a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5
Ablation (b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5
Ablation (c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0
Ablation (d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0
Ablation (e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0
Ablation (f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5
Ablation (g)	—	—	—	—	—	—	—	—
Ablation (h)	—	—	—	—	—	—	—	—

and representative visual states.

(g) Our RL + LLM-DexSafeDagger. This setting matches (f), except that the language model receives only the rollout statistics and no visual observations.

(h) Our RL + Grid-Search DexSafeDagger. This setting replaces the language/vision model in (f) with grid search over the intervention thresholds. The remaining teacher, student, success-value critic, and distillation components are held fixed.

To answer question (i), we compare (a) with (b) and (d) with (e). The DexTrAH-RGB-style RL recipe achieves lower performance than our RL recipe. We attribute this gap to the specialized tuning of the FABRICS controller for the original task setup, which transfers poorly to new hardware and dynamics.

We next address question (ii) by examining the distillation behavior in Fig. 3. Ablations (b), (c), (e), and (f) achieve comparable final lift success (left), consistent with the intervention policy’s focus on safety rather than task success. Unlike Vanilla DAgger ($\beta = 0$), the three intervention variants reduce teacher intervention as the student improves (middle). They reduce early unsafe episodes by approximately 80% relative to Vanilla DAgger and remain below 20% throughout distillation, with VLM-DexSafeDagger achieving the lowest

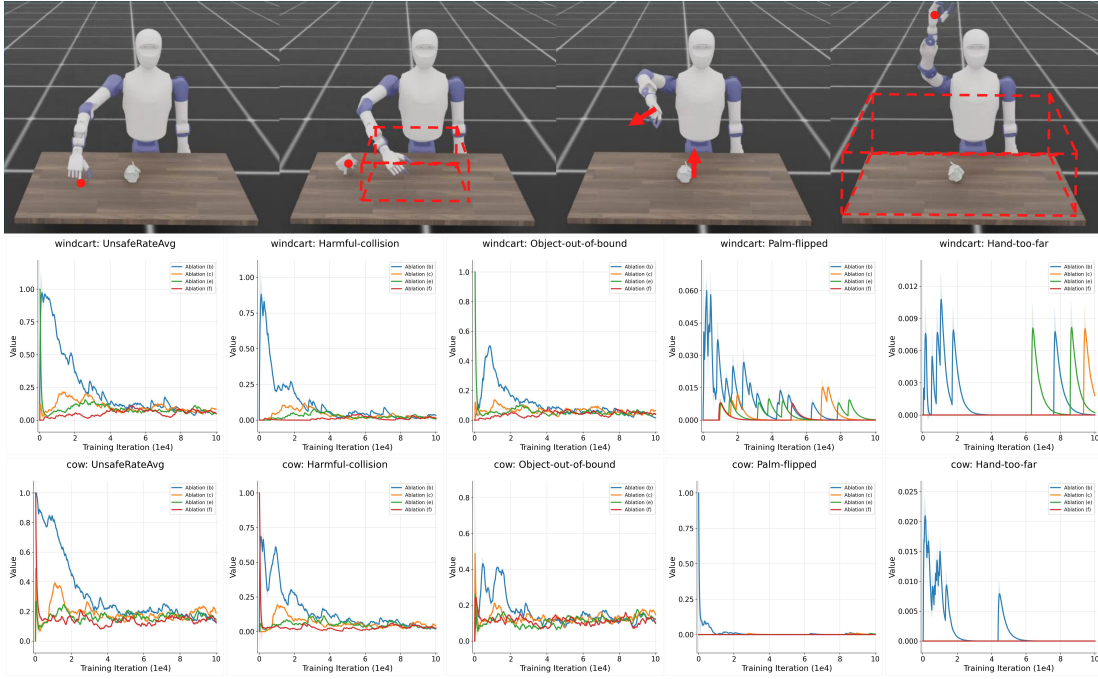


Fig. 4: **Per-object failure-mode decomposition during online distillation ablations.** We illustrate four task-specific failure modes in simulation (harmful collision, object escape, palm flipped, and hand too far) and report the aggregate unsafe rate together with its per-mode breakdown over training iterations. Results are shown for two objects (windcart and cow); lower is better in all plots. Curves correspond to four distillation variants: blue, Vanilla DAGger [10]; orange, Vanilla SafeDAGger [12]; green, DexSafeDAGger; and red, VLM-DexSafeDAGger.

rate overall (right).

We address question (iii) by comparing ablation (c), which uses disagreement alone, with ablation (e), which adds success-value prediction while retaining the same RL teacher and stationary intervention policy. The combined gate intervenes in low-predicted-success states missed by disagreement alone, reducing the average unsafe-episode rate by about 7.5 percentage points and limiting collision and object-escape failures without sacrificing task performance. Its unsafe rate approaches the teacher’s average ($\approx 12\%$) by the end of training and reaches a minimum of approximately 8%. Fig. 4 further shows that harmful collisions and object escape dominate unsafe episodes. Compared with Vanilla DAGger, all three intervention variants suppress both failure modes, particularly their early spikes, demonstrating that teacher intervention limits unsafe exploration during the initial distillation phase.

Finally, we address question (iv) by comparing ablations (e), (h), (g), and (f), which form a progression from stationary intervention, to grid-searched thresholds, to semantics-guided LLM adaptation, and finally to our VLM-guided recipe. **TODO: Add the quantitative comparison of ablations (e), (h), (g), and (f) when the evaluation results are available.** Across all evaluated objects, ablation (f) achieves an unsafe-episode rate that is lower than that of (e) by an average of 4 percentage points, while maintaining comparable lift success. This

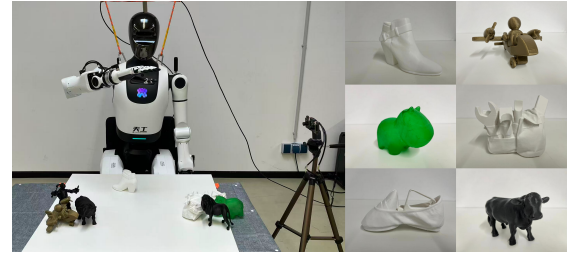


Fig. 5: **Real-world experimental setup and test objects.** Left: the single-arm manipulation platform used for deployment and evaluation. Right: examples of the physical objects used in the real-world lift experiments.

result shows that VLM-guided intervention-policy adaptation improves safety without sacrificing task performance.

D. Real-World Experimental Results

We evaluate the deployed student policy on 22 objects, comprising 16 seen objects and 6 unseen objects. Table IV reports a representative six-object subset and pooled results for the four seen and two unseen objects shown. The experiments exhibit a lower lift success rate and a higher unsafe episode rate than the teacher and student evaluation results in simulation, indicating that VLM-DexSafeDAGger does not fully eliminate the sim-to-real gap. Nevertheless, the resulting policy is sufficiently robust for real-world deployment.

TABLE IV: Representative six-object subset of the 22-object real-world evaluation (30 episodes per object). The last two columns report pooled results over the four seen and two unseen objects shown, respectively. **Unseen objects** are shaded.

Metric (%)	WindCart	Boot	ToolBox	Cow	Plane	Shoe	Seen avg. (4)	Unseen avg. (2)
Lift Success	86.7	86.7	80.0	86.7	66.7	60.0	85.0	63.3
Unsafe Rate Avg	13.3	20.0	20.0	26.7	40.0	40.0	20.0	40.0
Harmful collision	0.0	6.7	13.3	13.3	20.0	26.7	8.3	23.3
Object escape	13.3	13.3	6.7	13.3	20.0	13.3	11.7	16.7
Palm flipped	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

The unseen-object evaluation reveals a clear generalization gap: compared with the seen objects shown, the policy’s pooled lift success decreases from **85.0%** to **63.3%**, while its unsafe rate increases from **20.0%** to **40.0%** on the unseen objects shown. This degradation is primarily driven by increased harmful collisions (20.0% and 26.7% on Plane and Shoe), while object escape remains comparable across objects (≈ 6.7 –20.0%). No palm-flip or hand-too-far events are observed, suggesting that real-world failures are dominated by contact-rich interactions rather than gross pose instabilities. Overall, the results indicate that the remaining sim-to-real gap on unseen geometries is largely attributable to a contact-strategy mismatch, i.e., the policy fails to adapt its contact interactions to novel object shapes.

V. CONCLUSION AND FUTURE WORK

In this work, we propose VLM-DexSafeDagger, which integrates teacher–student disagreement with a Bellman success-value critic trained on safe-lift-success labels and uses a VLM to adapt the intervention policy. VLM-DexSafeDagger improves exploration safety during multi-teacher distillation, reducing the unsafe-episode rate by **80%** relative to the vanilla Dagger baseline during early training. The introduction of the Bellman critic reduces the average unsafe-episode rate by **7.5 percentage points** relative to the disagreement-only SafeDagger baseline. In addition, VLM-guided intervention policy adaptation yields consistent improvements across different visual contexts and unsafe scenarios, reducing the average unsafe-episode rate by a further **4 percentage points** relative to DexSafeDagger. The proposed method approaches and, in some cases, surpasses the teacher policy’s safety performance. Successful real-world deployment shows that the resulting student policy partially bridges the sim-to-real gap.

A clear limitation is that performance during distillation, as well as the final student policy, is largely constrained by teacher capability. Developing a safety-aware teacher training pipeline is a promising direction for future work to generate stronger teacher policies.

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